1 Purpose

This document introduces you to the main functions and features of DF Literature Monitor and provides the necessary instructions to use the DF Literature Monitor application.

1.1 Intended Audience

Job Classification	Description
Regulatory Specialist	People who ensure the product meets the regulatory standards and update with regulatory changes.
Drug Safety Specialist	Individuals who were responsible for monitoring and evaluating adverse drug reactions (ADRs) and ensuring drug safety.
Pharmacovigilance Officers	Professionals who oversee the collection, assessment, and reporting of safety data.
Medical Specialist	Experts in specialized fields (e.g., oncology, cardiology) who need in-depth knowledge of the latest advancements in their specialty.
Clinical Researchers	Scientists conducting clinical trials need to monitor the literature for emerging safety data.
Patients	Individuals seek information about the latest treatments and research developments relevant to their conditions.
Medical Affair Teams	Teams who were responsible for providing scientific and medical information to healthcare professionals and stakeholders.

1.2 Prerequisites

You should be familiar with your computer and its operating system.

For instance, you must know the operating system and specific commands for tasks like copying, deleting, and executing files.

2 Overview

DF Literature Monitor transforms the medical literature tracking process by automating the article search, listing, evaluation, and safety case initiation. The latest AI solution used in the DF Literature Monitor enhances productivity and ensures compliance. It reduces the time, effort and cost involved in tracking large volumes of local and global literature, clinical research, and scientific information. DF Literature Monitor improves the efficiency by streamlining the complexity of literature monitoring for Safety Vigilance through multi-level automation.

3 Literature Monitoring Process- Conceptual Description

The literature monitoring process within DF Literature Monitor follows a structured, end-to-end workflow designed to ensure regulatory compliance, operational efficiency, and high-quality safety surveillance. Each stage of the process integrates automated machine learning–driven capabilities with controlled human assessment to deliver consistent, traceable, and inspection-ready outputs. The following sections describe each step in detail.

3.1 Article Acquisition

Article acquisition is the first step in the literature monitoring workflow, enabling the intake of relevant scientific and medical publications. DF Lit Mon begins with automated retrieval of scientific publications using its AI-driven Intelligent Search Engine. This engine follows a structured, three-step process to identify and prepare PV-relevant literature with minimal manual intervention.

3.1.1 Multi-Database Article Ingestion

The platform integrates with 25+ global scientific and medical literature sources, including PubMed Central, BioMed Central, and Elsevier.
With minimal user input—such as product name, indication, or key adverse events—the system triggers AI-enabled searches across all configured databases. Using NLP and ML-based interpretation of medical language, the engine retrieves potentially relevant publications for safety monitoring.

3.1.2 Configurable Ingestion Frequency

DF Literature Monitor ingests new articles **daily**, ensuring timely availability.
Users may configure listing intervals to align with organizational or regulatory schedules, including:

- Daily
- Weekly
- Monthly
- Yearly

This supports compliance with requirements such as EMA's mandate for weekly literature review.

3.1.3 Automated Deduplication

As articles arrive from multiple databases, the system automatically identifies and removes duplicates by comparing:

- Titles
- Abstracts
- Author information
- Unique article identifiers

Only unique articles move forward into processing, ensuring a consolidated, clean corpus.

3.1.4 AI-Enabled Relevancy Ranking

Each ingested article is assigned a relevancy score using NLP-driven contextual understanding and drug–event associations. This ranking enables assessors and reviewers to focus on high-priority articles first, improving operational efficiency.

3.2 Pre-Processing pipeline

The pre-processing step standardizes the input documents, regardless of format or language, before they proceed to automated extraction.

3.2.1 Optical Character Recognition (OCR)

If the uploaded article is a scanned PDF, the system employs Optical Character Recognition (OCR) to convert the image-based text into a machine-readable format. This step digitizes the text for further processing.

3.2.2 Machine Translation

For non-English articles, the platform utilizes a machine translation model. Pre-trained models are deployed for 24 languages: Chinese, Croatian, Danish, Dutch, Estonian, Finnish, French, German, Hungarian, Indonesian, Italian, Japanese, Latvian, Lithuanian, Norwegian, Persian, Polish, Portuguese, Romanian, Russian, Slovak, Slovenian, Spanish, Swedish, ensuring accurate translation of the text into English.

3.2.3 Document Upload and Translation

DF Literature Monitor provides users with the capability to upload locally sourced literature articles in PDF format. The processing pipeline incorporates a combination of advanced machine learning (ML) techniques to handle various input formats and languages.

If the uploaded articles is a scanned PDF or non-English articles, it is first digitized, translated and pushed to the ML pipeline for MSI extraction.

3.3 Machine Learning- Based Extraction (NER and RE)

The ML pipeline extracts structured safety information to support automated triage.

3.3.1 Named Entity Recognition (NER)

NER is a specialized Natural Language Processing (NLP) technique used to identify and classify key entities within the text. DF Literature Monitor applies NER to detect and extract entities such as:

- Suspect products
- Patient information (age, age group)
- Adverse events

By automating entity extraction, NER significantly reduces manual annotation efforts and enhances processing efficiency. The NER algorithms are fine-tuned for high accuracy to ensure that critical information is accurately captured from literature articles.

3.3.2 Relationship Extraction (RE)

RE works in tandem with NER to establish and understand relationships between identified entities within the text. This technique is essential for identifying associations among drugs, patient information, and reported adverse events. RE enhances the identification of relevant

articles by establishing the association among extracted entities, thereby supporting comprehensive safety assessments.
By integrating NER and RE, DF Literature Monitor ensures precise and efficient extraction of MSI from literature articles, facilitating timely and accurate safety assessments.

3.4 MSI (Minimum Safety Information) Classification

MSI classification determines whether a literature article contains the regulatory-required elements for potential ICSR consideration:

- Identifiable patient
- Suspect product
- Adverse event
- Primary reporter

Article tagging

- Articles containing all four MSI components are automatically tagged “Yes – Potential AE.”
- Articles missing one or more components are tagged “No – Potential AE” with a structured explanation.

3.5 Workflow Models

3.5.1 One-step Workflow

In one-step workflow the assessor validates extracted information, and reviewer performs secondary verification ensuring accuracy.

- Accessor Review

Assessors conduct the first level of human quality review.

Actions

  - Validate or correct extracted MSI elements.
  - Confirm relevance to the safety surveillance scope.
  - Provide assessor comment.
  - Classify the article status prior to reviewer escalation.

- Reviewer Approval

Reviewers perform a secondary, independent verification step to ensure the accuracy and regulatory compliance of assessor decisions.

Actions

- Confirm MSI accuracy and overall assessment.
- Approve an article.
- Document rationale for approval or rejection.

3.5.2 Two-step Workflow

Associate performs dual-level review ensures compliance and consistency before final approval.

- Associate Workflow

An Associate does both the work of an assessor and reviewer in a single workflow. From checking whether the article is relevant, adding, updating, deleting the information as a part of the review process to conducting the final review while adding valid comments and additional relevancy if required.

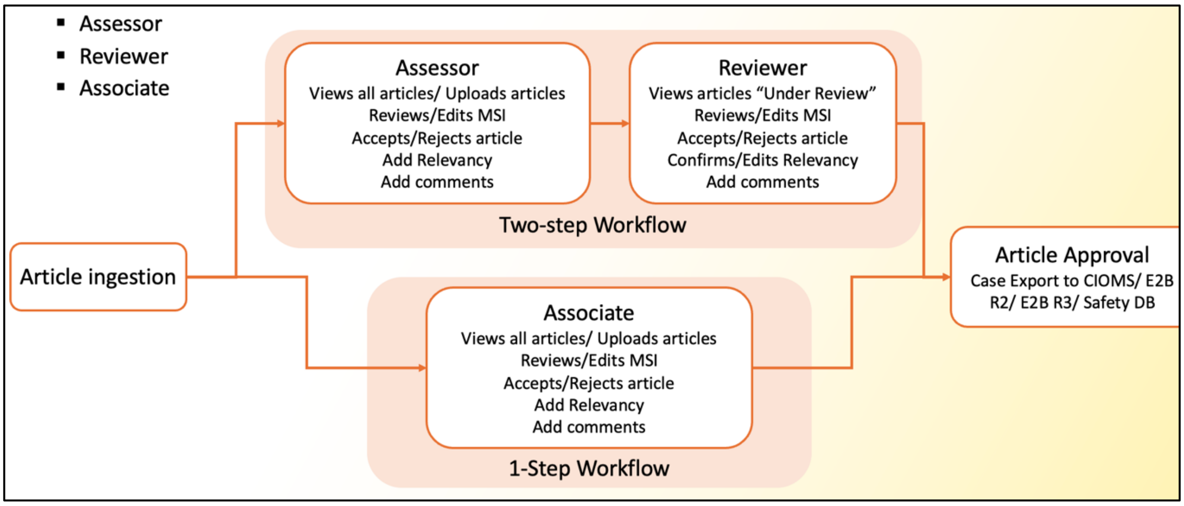

Figure 1 Picture showing the diagrammatic representation of one-step and two-step workflow in DF Literature Monitor.

3.6 Qualified Articles

Qualified articles are those that have completed both assessor and reviewer workflows. The Qualified Articles module provides a centralized interface for viewing all articles from which validated ICSRs, Aggregated Reports, or Signals have been identified. To support efficient navigation and data extraction, the section includes robust filtering and export functionalities designed to help users quickly locate relevant articles and generate offline records for regulatory or internal documentation.

A relevancy-based dropdown filter allows users to refine the displayed records based on the type of output generated from the article, including categories such as ICSRs, Aggregated Reports, or Signals.

For streamlined reporting and record-keeping, the module provides an Export function that allows users to download all qualified articles. This ensures that CRO teams can maintain offline datasets for audit purposes, regulatory submissions, or cross-functional review.

3.7 Reporting

The generated literature monitoring report provides a structured, end-to-end view of all scientific publications identified, screened, and assessed during the defined search period. It consolidates article details, product-safety information, and multi-level assessments to support regulatory-compliant pharmacovigilance activities.

The report captures each article retrieved from relevant scientific databases and journals, documenting key metadata such as publication details, authorship, and geographic information. It further identifies and extracts safety-relevant elements related to the suspect product, including adverse reactions, patient characteristics, seriousness, outcomes, and labeling status.

Overall, the report functions as a comprehensive evidence record that integrates scientific literature findings with structured safety evaluation, ensuring robust surveillance of emerging risks for the monitored product.

The DF Literature Monitor applies a structured, end-to-end workflow that combines automated machine-learning capabilities with human assessment to ensure regulatory-compliant, traceable, and inspection-ready literature monitoring. The process begins with **Article Acquisition**, where the system retrieves scientific publications through its AI-driven Intelligent Search Engine. This includes multi-database ingestion from 25+ global sources, configurable ingestion frequency aligned with regulatory schedules, automated deduplication, and AI-enabled relevancy ranking.

In the Pre-Processing Pipeline, uploaded articles undergo OCR for scanned PDFs, machine translation across 24 languages, and document normalization before being sent to the ML extraction pipeline.

The Machine Learning–Based Extraction stage uses NER to identify suspect products, patient details, and adverse events, while RE establishes relationships between extracted entities. Together, they ensure accurate and efficient extraction of Minimum Safety Information (MSI).

During MSI Classification, articles are evaluated for the four required MSI components—identifiable patient, suspect product, adverse event, and primary reporter—and are tagged accordingly as “Yes – Potential AE” or “No – Potential AE.”

The system supports two Workflow Models. In the One-Step Workflow, assessors perform validation and reviewers provide secondary verification. In the Two-Step Workflow, an associate completes both levels, from checking relevance to final review and commentary.

Qualified Articles are those that complete the full workflow and can be viewed, filtered by relevancy outputs (ICSRs, Aggregated Reports, Signals), and exported for regulatory submissions or audits.

Finally, Reporting consolidates all retrieved, screened, and assessed articles from the monitoring period. Reports capture publication details, extracted safety information, and assessment outcomes, forming a complete evidence record to support ongoing safety surveillance for the monitored product.

4 System Capabilities

The DF Literature Monitor platform incorporates several system-level capabilities designed to ensure operational efficiency, regulatory compliance, and secure management of users, roles, and platform-wide activities. These capabilities, spanning Role and Access Management, Alerts and Notifications, and Audit Trail tracking—support transparent workflow governance, streamlined collaboration, and enhanced oversight across all modules of the system.

4.1 Role and Access Management

It provides Role and Access Management framework that governs how users interact with the system based on assigned permissions. These controls ensure secure access to sensitive safety data and maintain structured workflow segregation across operational functions such as assessment, review, reporting, and administrative management.

Key Capabilities

- **Centralized User Management**

The User Management module enables administrators to view, edit, activate, or deactivate user accounts. User records include username, first name, last name, email, assigned role, and status indicators. Sorting and search functions allow administrators to quickly locate user profiles.

- **Role Assignment and Modification**

Administrators can modify a user’s role using the “Role Actions” column. The system presents a role-change pop-up, allowing selection from predefined roles. This capability ensures consistent permission alignment across evolving organizational structures.

- **Role Configuration Management**

The Role Configuration module defines all system-accessible modules and their associated features. Administrators can:

  - Edit existing roles
  - Assign or revoke permissions using checkbox-driven access controls
  - Add new roles and configure feature-level permissions

- **Standard Roles**

It includes four system-defined standard roles—Admin, Assessor, Reviewer, and Associate—which carry predefined, non-editable permissions that support segregation of duties in the literature monitoring workflow.

These functionalities collectively ensure secure, scalable, and easily auditable management of user-level access within the platform.

4.2 Alerts and Notifications

The platform includes an integrated Alerts and Notifications module to facilitate timely communication of key activities and workflow events. This functionality ensures that users remain informed of tasks or events requiring their attention, thereby supporting compliance-driven and timely case handling.

Key Capabilities

- **Module-Based Filtering**

Users can filter alerts based on specific modules (e.g., Project Configuration, Article Listing, Review Workflow) to view notifications relevant to their role and responsibilities.

- **Notification Controls**

Each alert can be toggled on or off using switch controls, allowing users to customize the type and volume of notifications they receive.

- **Configurable Notification Settings**

Administrators can: add recipient roles, assign individual recipient email addresses, enable or disable email or in-application notifications and configure module-specific alert rules.

Only the **Admin** role has permission to configure alert settings to ensure controlled management of communication rules across the system.

4.3 Audit Trail

DF Lit Mon maintains a comprehensive Audit Trail module designed to support regulatory compliance and ensure complete traceability of system activities. The audit trail captures user actions, changes, and updates across the system, enabling transparent oversight and effective documentation for audits, inspections, and internal governance.

Key Capabilities

- **Granular Audit Generation**

Admin users can generate audit reports for the following elements: Product, project, article and case. Each selection dynamically filters the audit records to reflect activity relevant to the chosen component.

- **Configurable Parameters**

The audit generation panel allows selection of: Product name, Project ID (if applicable), Article ID (if applicable) and Case ID (if applicable)

- **Submission and Reset Functionality**

Users may generate updated audit trail results based on new filters or reset the fields to clear all selections.

This module ensures that every significant configuration change, workflow update, and data modification is logged, accessible, and verifiable, thereby supporting industry-standard compliance expectations.

5 Getting Started

You must be familiar with your computer and its operating system.

For example**,** you must know the operating system and some specific commands, such as those required for copying, deleting, and executing files.

5.1 Logging In

- Open Google Chrome, Microsoft® Edge, or Microsoft® Internet Explorer browser.
- Navigate to the URL provided by your site administrator.
- Log in using single sign on with organization email ID

OR

- Log In using your DF Literature Monitor username and password.
- In the Username field, enter the username provided by your administrator. The username field is your e-mail ID.
- In the Password field, enter your user password provided by your administrator. The password field is case-sensitive. The password length should be a minimum of 8 Character length. The password should have one special character, one alphanumeric minimum.

  The following are the password policy requirements.

  - Must contain at least one Letter (a-z).
  - Must contain at least a number (0-9).
  - Must contain at least a special character [! @#$_-=*()^&].

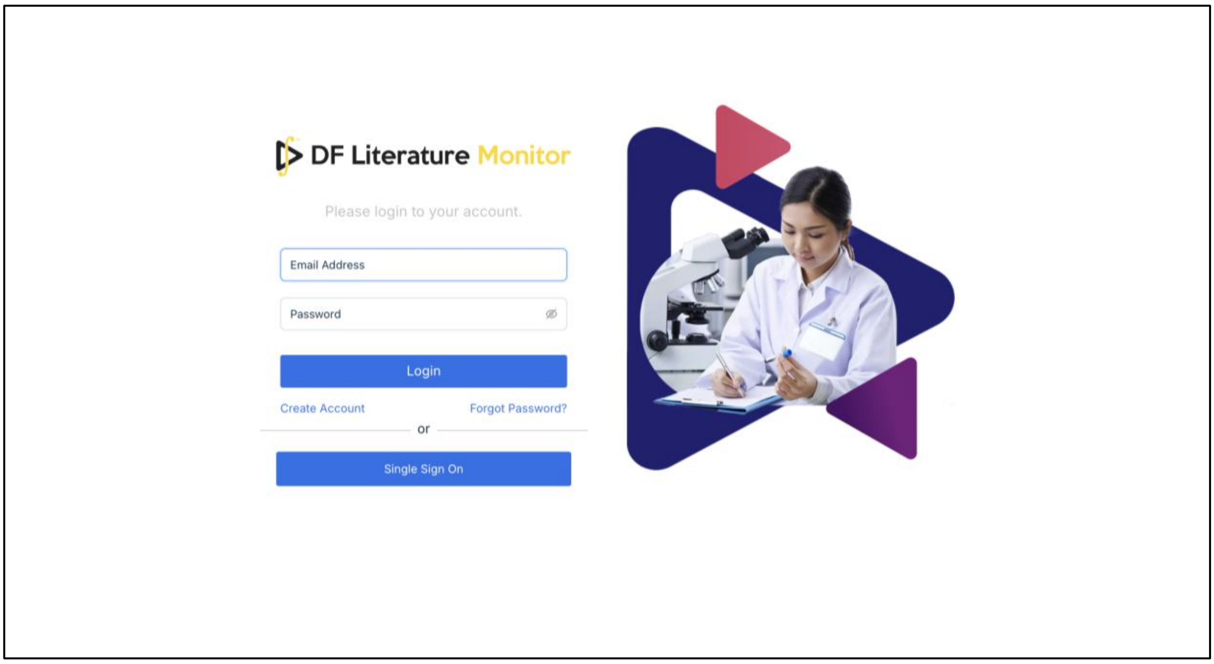

Figure 2 Picture displays DF Literature Monitor's Login screen.

- Click **Login** or press Enter. It will display the DF Literature Monitor Dashboard.

5.2 Logging out

- Click the drop-down next to the username in the upper right corner of the home screen. Click the **Logout** button.

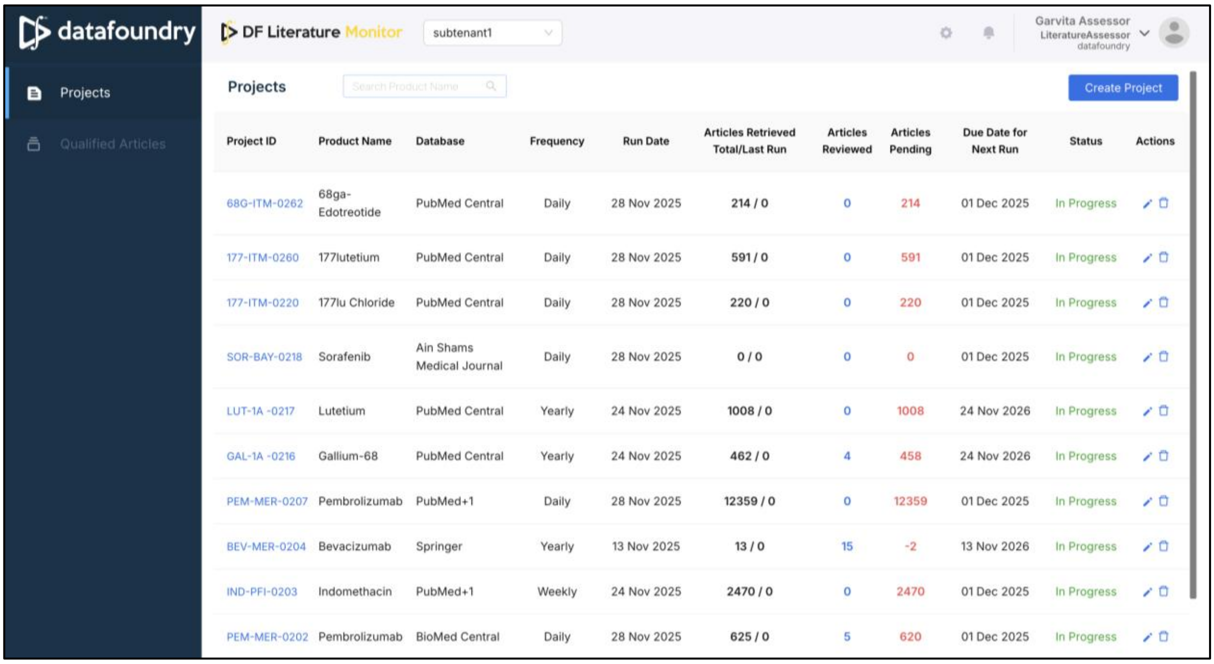

Figure 3 Picture displays home screen.

- The user will be successfully logged out from the DF Literature Monitor application.

6 Home Screen

The DF Literature Monitor home screen provides easy and quick access to modules such as Projects and Qualified Articles.

Menus:

Menus on the Home page (Left Panel):

- Projects
- Qualified Articles

Upper Right Corner Buttons and Links:

The upper-right corner Buttons and Links allow you to access your profile and the DF Literature Monitor application's setting information.

7 Projects

The Project module showcases all the active projects to ensure easy access to all the latest safety information for configured products.

Click on the Projects module from the left-navigation panel, it displays the project listing screen.

7.1 Create Project

- On the Project module screen, click the "Create Project" button located at the top right corner. Displays the Create Project screen.

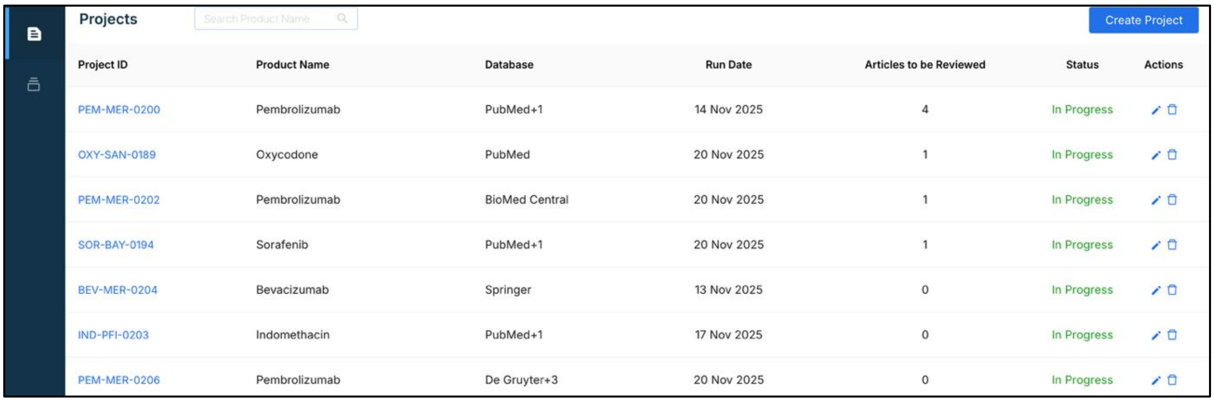

Figure 4 Picture displays Projects Listing screen.

- On the Create Project screen, fill in the mandatory fields. The Mandatory fields are:
    - Select Data Source – Multi-select.
    - Product Name – Single select.
    - Brand Names – Multi-select.
    - Select frequency – Single-select.
    - Select date – Single-select.